

Day 20 - Lecture 20.18: Binomial Distribution

 Python 3.10+

 notes.pdf

 Language: English | [हिंदी / Hinglish](#) |  [Back to Lecture 20 Module](#)

1. Overview & Bernoulli Trial Foundation

The **Binomial Distribution** models the exact number of successes k in a fixed sequence of n independent and identically distributed (i.i.d.) **Bernoulli trials**, each with a constant probability of success p .



Necessary Assumptions:

1. Fixed number of trials n .
2. Binary outcomes per trial (Success vs Failure).
3. Constant probability of success p across all trials.
4. Independent trials.

2. Mathematical Formalism

Probability Mass Function (PMF):

The probability of observing exactly k successes in n trials:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} \quad \text{for } k \in 0, 1, \dots, n$$

where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ accounts for all distinct permutation orders.

Mathematical Derivation of Moments:

Expected Value (Mean μ):

Since $X = \sum_{i=1}^n I_i$ where $I_i \sim \text{Bernoulli}(p)$ and $\mathbb{E}[I_i] = p$:

$$\mathbb{E}[X] = \mu = \sum_{i=1}^n \mathbb{E}[I_i] = n \cdot p$$

Variance (σ^2):

By independence, covariances vanish:

$$\text{Var}(X) = \sigma^2 = \sum_{i=1}^n \text{Var}(I_i) = n \cdot p \cdot (1 - p)$$

$$\sigma = \sqrt{n \cdot p \cdot (1 - p)}$$

3. Real-World Demonstrative Example

A customer conversion model evaluates $n = 5$ website visitors with independent conversion probability $p = 0.50$. What is the probability that exactly $k = 3$ visitors convert?

$$\boxed{P(X = 3) = \binom{5}{3} (0.5)^3 (0.5)^2 = 10 \cdot (0.125) \cdot (0.25) = \frac{10}{32} = \frac{5}{16} = 0.3125 \text{ quad } (31.25\%)}$$

4. Python Implementation

```
from scipy.stats import binom
import numpy as np

n = 5
k = 3
p = 0.5

# 1. PMF calculation
prob_pmf = binom.pmf(k, n, p)
print(f"Scipy PMF P(X = {k} | n={n}, p={p}): {prob_pmf:.4f} (Exact: 5/16 = {5/16:.4f})")

# 2. Cumulative Distribution Function (CDF): P(X <= 3)
prob_cdf = binom.cdf(k, n, p)
print(f"Cumulative P(X <= {k}): {prob_cdf:.4f}")

# 3. Moments verification
mean_ana = n * p
var_ana = n * p * (1 - p)
print(f"Analytical Mean = {mean_ana:.2f}, Variance = {var_ana:.4f}")
```

5. Key Takeaways & Interview Points

- **Maximum Variance at $p = 0.5$:** $\text{Var}(X)$ is maximized when outcomes are most uncertain ($p = 0.5$).
- **Normal Approximation:** When n is large and $np \geq 5, n(1 - p) \geq 5$, the Binomial distribution converges smoothly to a Gaussian $\mathcal{N}(np, np(1 - p))$.